

Operating Systems Fundamentals

Operating System

The most important piece of software

- Controls all the computer resources
- Manages the execution of application programs
- Shields programmers from the complexity of the hardware
- Provides users with command line (CLI) and graphical (GUI) interfaces



The UNIX Operating System



- Written by Ken Thompson and Dennis Ritchie at AT&T Bell Labs in 1972
- Was the first widely-used portable operating system
- Allows access by multiple users (Multuser)
- Concurrently performs multiple tasks (Multitask)
- Two major versions from which all others derive, System V from AT&T and BSD from the University of Berkeley
- IEEE developed a standard for UNIX called POSIX to make it possible to write programs that could run on any UNIX system

The Free Software Foundation



In the 70s, programmers used to share their code freely among each other. In the 80s, a new trend emerged where companies developing the code refused to share it with their clients

In response to this trend, Richard Stallman (RMS) formed the Free Software Foundation (FSF). The plan was to develop a free OS called GNU, which would be compatible with the proprietary Unix operating systems

GNU OS



The components required for kernel development were written first: an editor (emacs), a C compiler (gcc), a shell (bash), libraries and all associated generic Unix tools.

The GNU kernel is called Hurd and it uses a revolutionary server-client architecture built on top of a microkernel. The Hurd is still not considered suitable for production environments as of today...

Linux Torvalds



*From: torvalds@klaava.Helsinki.FI (Linus Benedict Torvalds)
Newsgroups: comp.os.minix
Subject: What would you like to see most in minix?
Summary: small poll for my new operating system
Message-ID: <1991Aug25.205708.9541@klaava.Helsinki.FI>
Date: 25 Aug 91 20:57:08 GMT
Organization: University of Helsinki*

Hello everybody out there using minix –

I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones. This has been brewing since april, and is starting to get ready. I'd like any feedback on things people like/dislike in minix, as my OS resembles it somewhat (same physical layout of the file-system (due to practical reasons) among other things).

I've currently ported bash(1.08) and gcc(1.40), and things seem to work. This implies that I'll get something practical within a few months, and I'd like to know what features most people would want. Any suggestions are welcome, but I won't promise I'll implement them ☺

Linus (torvalds@kruuna.helsinki.fi)

PS. Yes – it's free of any minix code, and it has a multi-threaded fs. It is NOT protable (uses 386 task switching etc), and it probably never will support anything other than AT-harddisks, as that's all I have :-).

GNU/Linux



The Linux kernel became the most popular adoption of the GNU software. A number of distributors provided a complete OS by putting together the Linux kernel, GNU tools and libraries, package managers, installers and other applications.



Nowadays Linux is pretty much everywhere: computers, smartphones, cars and embedded devices.
It basically powers the world...

BSD Family

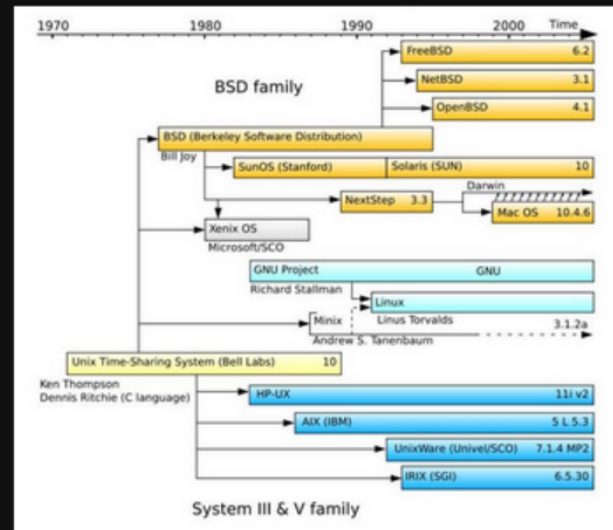


Berkeley Software Distribution (BSD) is a Unix operating system derivative developed and distributed by the University of Berkeley

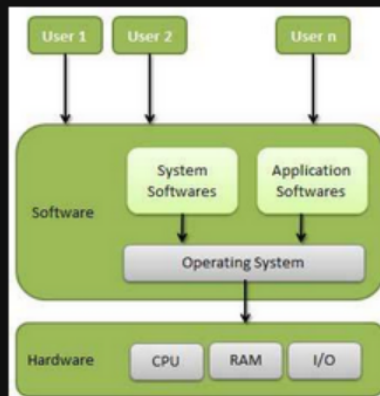
BSD has been the base of a large number of operating systems:

- FreeBSD - Known for its performance and rock solid stability
- OpenBSD - Known for being the most secure operating system
- NetBSD - Known for being the most portable operating system
- Darwin - FreeBSD fork and the base for Apple's OSX, iOS, watchOS and tvOS

Unix Family Tree



Operating System Overview



Consists of:

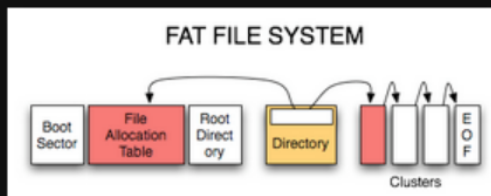
- Kernel
- Libraries and Utilities

Performs:

- Device management
- Filesystem management
- Process management
- Memory management

Filesystem

Defines the structure and the rules used to read, write and maintain data on a disk. Application software usually does not write directly to disk, but goes through the OS filesystem calls

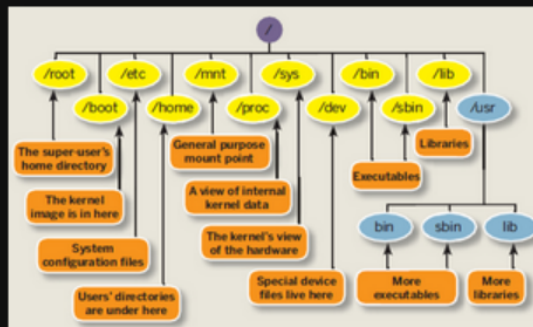


Common file system types include:

- Ext2 - Linux filesystems
- Ext3/Ext4 - Linux filesystem with journaling capabilities
- HFS/HFS+/AFS - Filesystems used by Apple computers
- FAT12/16/32 - Simple legacy filesystems developed by Microsoft
- NTFS - Proprietary Microsoft filesystem

Filesystem Hierarchy

The Filesystem Hierarchy Standard (FHS) defines the directory structure and directory contents in Unix-like operating systems

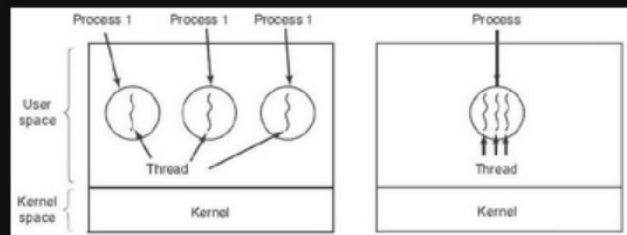


Different Operating Systems follow this standard to different levels of compliance

Processes and Threads

A process is a program in execution in memory. Any program executed creates a process. A program can be a command, a shell script, or any binary executable or any application.

A thread is a sequence of instructions and a lightweight alternative to processes. Threads within the same process share the same memory address space.

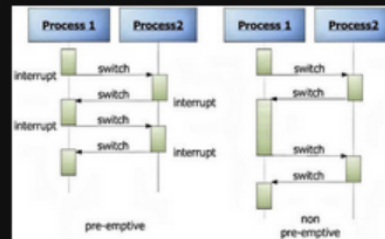


Scheduler

The Scheduler is an operating system module that selects and allocates CPU time to processes in memory that are ready to execute

The Scheduling algorithms can be divided into two categories:

- Nonpreemptive Scheduling - once a process has been given the CPU, the CPU cannot be taken away from that process and it has to voluntarily give the CPU back to the OS
- Preemptive Scheduling - once a process has been given the CPU, the OS can take it back at any time



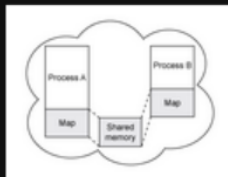
Preemptive scheduling is commonly used in real-time operating systems, such as VxWorks, QNX, Montavista Linux Symbian and many others. They are quite popular in audio, video and telecommunication applications

Interprocess Communication

Processes frequently need to communicate with other processes.

Sure, we can use Files, but it is slow and may require file locks. Such functionality is better provided by the kernel with IPC mechanisms:

- Message Queues - message exchange between processes
- Semaphores - synchronization between processes
- Signals - notification mechanism between processes
- Pipes/Sockets - connect streams of data
- Shared Memory - fast exchange of data between processes



{ Hands On }

Check Running Processes

Secure Shell Connection

Cryptographic network protocol that allows remote login.
Authenticate on a remote machine and run commands remotely, as if we were there!

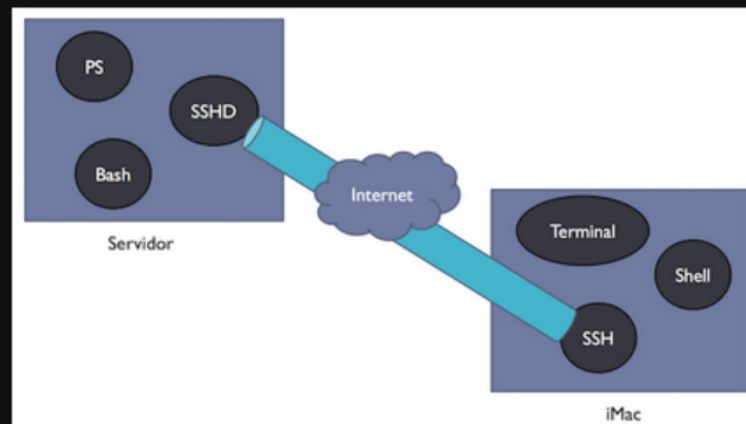
ssh - Authenticate using SSH protocol

scp - Copy files using SSH protocol

Asymmetric cryptography for usage without passwords

{ Hands On }

Remote Processes



Memory Management

Physical Memory is the amount of RAM available to the OS

Virtual Memory is a combination of RAM and disk space that running processes can use. Virtual memory makes the system appear to have more memory than it actually has by sharing it between competing processes as they need it.

Swap space is the portion of virtual memory that is on the hard disk, used when RAM is full

- Allocate memory to processes
- Free previously allocated memory
- Map virtual memory to physical memory
- Memory swap

{ Hands On }

Memory Usage and Statistics